

ActuaryGPT — Underwriting Report

Confidential actuarial risk assessment & premium recommendation report.

1. Customer & Policy Profile

Age (Normalized)	0.55	Gender	Male
Height (cm)	172.0	Weight (kg)	88.0
BMI	29.7	Product Info 2	D2
Occupation	Sales Director	Annual Income	\$1,500,000.00
Insurance Type	Life	Coverage Amount	\$9,000,000.00

2. Risk Analysis & Premium Pricing

Predicted Risk Class	Class 1 of 8
Risk Category	Low Risk
Recommended Annual Premium	\$10,000.00

3. AI Actuarial Analysis

Actuarial Report: Insurance Application Evaluation

Applicant Details:

* **Age:** 0.55 years
* **Height:** 172.0 cm
* **Weight:** 88.0 kg
* **BMI:** 29.7 (Overweight/Borderline Obese)
* **Occupation:** Sales Director
* **Income:** \$1,500,000.0
* **Smoker:** Yes
* **Previous Claims:** 2
* **Family History:** Present (1)
* **Insurance Type:** Life
* **Coverage Amount:** \$9,000,000.0
* **Exercise:** Yes
* **Alcohol:** Moderate (2)
* **Gender:** Male
* **Product Info:** D2

Machine Learning Classifier Output:

* **Predicted Risk Class:** 1 (Lowest Risk)
* **Prediction Probability/Confidence:** 68.0%
* **Calculated Annual Premium:** \$10,000.00

1. Risk Assessment

* **Age:** The reported age of 0.55 years (approximately 6-7 months old) is highly anomalous and logically inconsistent with other applicant details such as 'Sales Director' occupation, \$1.5M income, previous claims, exercise, and alcohol consumption. This is the **most significant risk factor/data discrepancy** and severely impacts the credibility of the entire application data.

* **BMI (29.7):** Falls within the 'Overweight' category (borderline 'Obese' as BMI ≥ 30). This is a moderate health risk factor that can contribute to various health complications.

* **Smoking Status (1):** Applicant is a smoker, which is a major, well-established risk factor for life insurance due to increased mortality and morbidity rates across numerous diseases. This alone typically places an applicant in a higher risk category.

* **Occupation (Sales Director):** Generally considered a standard risk occupation. The high income suggests a legitimate need for substantial coverage.

* **Previous Claims (2):** Two previous claims are a significant red flag, indicating a history of insurable events. The nature and severity of these claims would require detailed investigation to assess future risk.

* **Family History (1):** Indicates a history of adverse medical conditions within the family, which is a moderate risk factor for hereditary conditions.

* **Coverage Amount (\$9,000,000.0):** This is a high coverage amount, but appears to be within reasonable multiples (6x) of the reported high income (\$1,500,000.0) for a Sales Director.

* **Exercise (1) & Alcohol (2):** Positive lifestyle indicators (regular exercise, moderate alcohol) are somewhat

offset by the smoking status and high BMI.

* **Gender (Male):** Statistically, males generally have slightly higher mortality rates than females.

2. Comparative Analysis

The ML classifier identified three historical approved cases (POL-50578, POL-68475, POL-8638) with 100% similarity, all assigned Risk Class 1 and a premium of \$10,000.00. This implies that, based on the features used by the similarity engine, the current application is an exact match to these approved precedents.

Key Observations:

* The exact match in similarity suggests that, based on the available input features *as processed by the RAG system*, the applicant aligns perfectly with previously underwritten and approved low-risk profiles.

* **However, this 100% similarity is highly questionable and potentially misleading due to the critical anomaly in the applicant's age.** It is extremely improbable that the historical cases also shared an age of 0.55 years while holding positions like 'Sales Director'. This suggests that either the 'age' field was not a primary feature for similarity matching in the RAG context or that the historical cases themselves contained similar data entry errors, which is unlikely for approved policies.

3. Anomaly & Fraud Detection

* **Critical Age Anomaly:** The most glaring anomaly is the applicant's age of 0.55 years. This is physically impossible for an individual with an occupation of 'Sales Director', an income of \$1.5M, previous claims, and lifestyle habits (exercise, alcohol). This points strongly to a **data entry error, a placeholder value, or potentially an attempt at data manipulation.** This specific data point invalidates the entire profile as presented.

* **High Previous Claims (2):** While not impossible, two previous claims warrant thorough investigation into their nature, dates, and payout amounts, especially for an applicant predicted as Risk Class 1.

* **Smoker with Low Risk Class:** Being a smoker typically elevates an applicant to a higher risk category. A classification of Risk Class 1 for a smoker, especially with other potential health indicators (high BMI, family history, previous claims), is unusual and requires scrutiny.

* **BMI (29.7):** While not an extreme outlier, it is at the higher end and contributes to overall health risk, which seems to be underrepresented by a Risk Class 1 assignment.

4. Premium Recommendation

The ML model proposes an annual premium of \$10,000.00, corresponding to a predicted Risk Class 1 with a confidence of 68.0%. This premium is consistent with the top similar historical cases.

Justification:

* If the applicant's data *excluding the age anomaly* were accurate and truly represented a Risk Class 1 individual despite being a smoker with a high BMI and two previous claims, then the premium of \$10,000.00 would be justifiable based on the model's output and historical precedents.

* **However, given the fundamental and critical anomaly in the 'age' field, the validity of the ML prediction and subsequently the calculated premium of \$10,000.00 is severely compromised.** The current premium cannot be recommended without a thorough re-evaluation of the applicant's true age and its impact on all other risk factors. The 68.0% confidence, while moderate, does not alleviate the concern raised by the age anomaly.

5. Underwriting Decision Recommendation

****Refer for Urgent Manual Review.****

A direct approval or rejection is not feasible at this stage due to the critical data inconsistency.

****Required Actions for Manual Review:****

1. ****Verify Applicant's Age:**** This is paramount. The applicant's actual date of birth and current age must be immediately verified and corrected.
2. ****Re-evaluate Risk Profile:**** Once the age is corrected, the entire risk profile must be re-assessed, considering the combination of smoking status, high BMI, family history, and previous claims. It is highly probable that the applicant, assuming an adult working age, would fall into a higher risk class than 1.
3. ****Investigate Previous Claims:**** Obtain detailed information on the nature, dates, and outcomes of the two previous claims.
4. ****Confirm Smoking Status:**** Re-verify smoking status with medical records or cotinine tests.

The current ML output and historical comparison are rendered unreliable by the fundamental data integrity issue concerning the applicant's age. It is highly likely that, with a corrected age, the applicant would require a significantly higher premium or potentially be declined depending on the severity of the corrected risk factors.

4. Sign-off & Recommendation

Based on the predictive model output and AI underwriting guidelines, the premium recommendation is formally calculated at **\$10,000.00** per annum. The underwriting team should perform secondary manual verification if the risk class exceeds Class 5.